The Guo–Liu bound for weighted Hermite polynomials

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MathDB 346243; Guo–Liu, Conjecture 1, equation (29) in the published paper [1], and equation (30) in arXiv:2004.10441v1.

Task 1. Prove the original bound for every degree

Use the physicists' normalization

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

Write

$$N_n = 2n+1, \quad C_n = \frac{2^n \Gamma((n+1)/2)}{\sqrt{\pi}}, \quad g_n(x) = (1 - x^2/N_n)^{-1/2}, \quad x_n = \sqrt{N_n}\left(1 - \frac{\pi}{2N_n}\right).$$

In particular $0 < x_n < \sqrt{N_n}$ for every $n \geq 1$.

Theorem 1. *For every integer $n \geq 1$ and every real $x > 0$,*

$$|H_n(x)e^{-x^2}| \leq C_n g_n(\min\{x, x_n\}) e^{-x^2/2}. \tag{1}$$

Put $y_n(x) = H_n(x)e^{-x^2/2}$. The two standard identities needed below follow directly from the Hermite differential equation and $H'_{n+1} = 2(n+1)H_n$:

$$y_n'' + (N_n - x^2)y_n = 0, \qquad y'_{n+1} + xy_{n+1} = 2(n+1)y_n. \tag{2}$$

We also use the classical global Hermite inequality

$$|y_n(x)| \leq \sqrt{2^n n!} \qquad (x \in \mathbb{R}), \tag{3}$$

in precisely the normalization recorded in DLMF 18.14.9 [2]. This known inequality is an input, not a new assertion of this note.

The local bound, including the odd degrees

Lemma 1. *For every integer $n \geq 1$ and $0 \leq x < \sqrt{N_n}$,*

$$|y_n(x)| \leq C_n(1 - x^2/N_n)^{-1/2}.$$

Proof. Define the oscillator energy

$$E_n(x) = y_n'(x)^2 + (N_n - x^2)y_n(x)^2.$$

Equation (2) gives $E_n'(x) = -2xy_n(x)^2 \le 0$ for $x \ge 0$. For even n, $y_n'(0) = 0$ and $|y_n(0)| = n!/(n/2)! = C_n$, so

$$(N_n - x^2)y_n(x)^2 \le E_n(x) \le N_n C_n^2.$$

This proves the assertion for even degrees.

For odd n, apply the even-degree energy estimate to $v = y_{n+1}$. Set $M = N_n + 2$ and $q = M - x^2 > 0$. Weighted Cauchy–Schwarz and (2) yield

$$\begin{aligned} 4(n+1)^2 y_n(x)^2 &= (v'(x) + xv(x))^2 \\ &\le (v'(x)^2 + qv(x)^2)(1 + x^2/q) \\ &\le MC_{n+1}^2 \frac{M}{M - x^2}. \end{aligned}$$

Thus, in fact, throughout $0 \le x < \sqrt{M}$,

$$|y_n(x)| \le D_n(1 - x^2/M)^{-1/2}, \qquad D_n = \frac{C_{n+1}\sqrt{M}}{2(n+1)}. \tag{4}$$

We claim $D_n < C_n$. Write $n = 2m - 1$, $m \ge 1$. The gamma duplication formula gives

$$\left(\frac{D_n}{C_n}\right)^2 = Q_m := \left(m + \frac{1}{4}\right)\left(\frac{\Gamma(m + 1/2)}{\Gamma(m+1)}\right)^2.$$

Its consecutive ratio satisfies the exact identity

$$\frac{Q_{m+1}}{Q_m} = 1 + \frac{1}{4(m+1)^2(4m+1)} > 1.$$

Stirling's formula gives $Q_m \longrightarrow 1$. A strictly increasing sequence with limit 1 has every term less than 1, proving the claim. Finally, for $x^2 < N_n < M$,

$$(1 - x^2/M)^{-1/2} \le (1 - x^2/N_n)^{-1/2}.$$

Together with (4), this completes the odd case. □

The capped region for degrees at least three

Let $A_n = C_n g_n(x_n)$. Its exact value satisfies

$$A_n^2 = C_n^2 \frac{N_n^2}{\pi(N_n - \pi/4)}.$$

We show $A_n^2 > 2^n n!$ for all $n \ge 3$. Define

$$r_n = \frac{C_n^2}{2^n n!} = \frac{\Gamma((n+1)/2)}{\sqrt{\pi}\,\Gamma(n/2+1)}, \qquad U_n = \frac{A_n^2}{2^n n!} = \frac{r_n N_n^2}{\pi(N_n - \pi/4)}.$$

Since $r_{n+2}/r_n = (n+1)/(n+2)$, putting $N = N_n$ and $a = \pi/4 < 1$ gives

$$\frac{U_{n+2}}{U_n} = \frac{(N+1)(N+4)^2(N-a)}{(N+3)N^2(N+4-a)} > 1.$$

For completeness, the numerator minus the denominator is

$$\begin{aligned} &2\{N^3+(6-3a)N^2+(8-12a)N-8a\} \\ &\qquad > 2(N^3+3N^2-4N-8) > 0 \qquad (N \geq 3). \end{aligned}$$

The last inequality follows from $N^3 - 4N > 0$ and $3N^2 - 8 > 0$. The elementary bound $\pi < 22/7$, and the monotonicity of the relevant denominators on this interval, give the two base cases

$$U_3 = \frac{784}{3\pi^2(28-\pi)} > \frac{33614}{31581} > 1, \qquad U_4 = \frac{243}{2\pi(36-\pi)} > \frac{11907}{10120} > 1.$$

Induction in each parity proves the claim. Therefore (3) implies $|y_n(x)| < A_n$ for every real x when $n \geq 3$.

The two small degrees

Direct differentiation gives

$$\sup_{x\geq 0} |y_1(x)|^2 = 4/e, \qquad \sup_{x\geq 0} |y_2(x)|^2 = \max\{4, 64e^{-5/2}\}.$$

Here $H_1(x) = 2x$ and $H_2(x) = 4x^2 - 2$; the respective nonzero critical points are $x = 1$ and $x = \sqrt{5/2}$. Again using $\pi < 22/7$,

$$A_1^2 = \frac{144}{\pi^2(12-\pi)} > \frac{6174}{3751} > \frac{8}{5} > \frac{4}{e},$$

because $e > 5/2$. Similarly,

$$A_2^2 = \frac{400}{\pi(20-\pi)} > \frac{4900}{649} > \frac{15}{2} > \max\{4, 64e^{-5/2}\}.$$

For the last step it suffices that $e^{5/2} > 1 + 5/2 + 25/8 + 125/48 = 443/48 > 9$, since $64/9 < 15/2$. All comparisons are exact, and give $|y_n(x)| < A_n$ also for $n = 1, 2$.

Completion of the proof. For $0 < x \leq x_n$, apply Lemma 1. For $x \geq x_n$, apply the global cap just established for every degree. Multiplication by $e^{-x^2/2}$ proves (1) in both regions. □

Scope. The proof covers every integer degree, every positive real argument, and the exact cutoff prescribed in the source. It also proves the stronger odd-degree local estimate (4). Exact symbolic checks accompany the paper; finite numerical samples are not used to justify the universal inequality. The proof was reviewed by the authoring agent; independent peer review is not claimed.

Task 2. Optimize the common cutoff constant

Replace the source's $\pi/2$ by a constant $c \in (0, 3)$ in $x_n(c) = \sqrt{N_n}(1 - c/N_n)$. Determine the largest c for which the capped bound holds for all $n \geq 1$. The degree-one maximum gives an explicit necessary restriction; determine whether that restriction is also sufficient for every higher degree.

Task 3. Optimize the local envelope

For each degree, determine the least exponent θ_n such that $|y_n(x)| \le C_n(1-x^2/N_n)^{-\theta_n}$ throughout $0 \le x < \sqrt{N_n}$. Study its large-degree limit, its parity dependence, and the locations at which the optimal bound is attained. Compare the result with the stronger odd-degree estimate in this proof.

Task 4. Improve the summation remainder quantitatively

Insert the minimum of the local energy bound and the classical global bound into the one- and two-dimensional remainder integrals of Guo–Liu, instead of using the constant A_n throughout. Derive explicit computable remainder bounds and optimize the choices of truncation rank and Fourier cutoff for a prescribed tolerance.

Task 5. Extend the parity transfer to generalized Hermite functions

For the generalized Hermite family associated with $|x|^{2\mu}e^{-x^2}$, $\mu > -1/2$, determine which adjacent-degree differential identities yield an analogous local bound with explicit gamma constants. Establish the admissible parameter range and a uniform transition to a finite global cap, accounting for the singular term at the origin in the transformed differential equation.

References

[1] J. Guo and Y. Liu, *A modified Euler–Maclaurin formula in 1D and 2D with applications in statistical physics*, Commun. Theor. Phys. **73** (2021), 075002, Conjecture 1, equation (29), p. 8. doi:10.1088/1572-9494/abf1dd. Preprint: arXiv:2004.10441v1.

[2] NIST Digital Library of Mathematical Functions, equation 18.14.9, global bound for Hermite polynomials; accessed 17 September 2026.